Department of Computer Science, Electrical and Space Engineering
D7043E Advanced Data Mining
Task 1 - MID term project
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Edge-Ready ECG Classification

Project using CRISP-DM

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Table of Contents

1. Business/Application Understanding **1**
2. Data Understanding **2**
2.1 Dataset Description 2
2.2 Target Classes 2
2.2.1 ECG Signal and Heartbeat Representation 3
2.3 Exploratory Data Analysis (EDA) 3
2.3.1 Example ECG Signal 3
2.3.2 Class Distribution 6
2.3.3 Record-Level and Subject Distribution Variability 7
2.3.4 Signal Amplitude Distribution 8
2.3.5 Examples of noisy ECG 8
2.4 Data Quality 9
2.4.1 Missing Data 9
2.5 Leakage Analysis 10
3. Data Preparation **10**
3.1 Record Selection 10
3.2 Channel Selection 11
3.3 Beat Extraction 11
3.4 Signal Preprocessing 12
3.5 Label preparation 13
3.6 Class Imbalance 14
3.7 Augmentation 14
3.8 Final Dataset 14
3.9 Reproducibility 14
Bibliography **14**

1. Business Understanding

1.1 Application Scenario

Cardiac arrhythmias are abnormalities in the heart rhythm that can cause the heart to beat too quickly, too slowly, or irregularly (Nagpal et al., 2024). These arrhythmias are among the leading causes of morbidity and mortality worldwide, and their timely detection depends on correctly interpreting the electrocardiogram (ECG). Interpreting ECG manually is slow, expensive, and depends on expert cardiologists (Rafie et al., 2021). A single 24-hour Holter recording contains roughly 100,000 heartbeats which is far too many for a human to annotate reliably. The core problem is automatic detection and classification of the individual heartbeats so that clinically meaningful arrhythmias are highlighted without manual interpretation. This project addresses that problem by training a compact neural network that classifies the ECG segments of heartbeats, and by preparing this network to run on a resource-constrained Edge-AI microcontroller called MAX780002.

Automatic ECG classification could support analysis in wearable or portable ECG monitors, Holter monitoring, and remote or continuous cardiac monitoring. In these scenarios, a classifier could help flag beats for review by a clinician. However, it would not replace clinical interpretation or independently establish a diagnosis but any mismatch between data volume and review capacity is precisely what automated classification is meant to close (Tan et al., 2005).

Running inference locally on a device such as the MAX78002 could reduce dependence on sending continuous ECG data to a remote server, limit communication overhead, and support faster or more reliable operation when there is limitation on connectivity (Scrugli et al., 2021). Since no physical kit is required, this project will demonstrate compatibility through software workflow, including quantization and synthesis, and evaluate the model from the software aspect rather than performance measured on a physical device.

> *We imagine potential users who are cardiologists could use this as a triage in order to label heart beats and highlight events. Hospitals and similar industries can benefit with continuous monitoring and automated alarms. Emergency medical services could use it for rapid on-scene triage. Other potential users include researchers and developers requiring scalable ECG analysis. Furthermore, patients could be end users who could utilize a wearable or portable device for regular monitoring.*

1.2 Data-mining objective

The machine-learning problem is a supervised pattern classification task: learn a mapping from a fixed-length ECG beat representation to a discrete diagnostic label, using a labeled training set of beats whose true class is known. From the perspective of data-mining, this is a supervised classification problem: given a representation of a heartbeat, assign it to a discrete class label such as normal or one of several arrhythmia types. The main difficulty is not merely accuracy but it is to uphold accuracy on a resource-constrained device (Moody & Mark, 2005).

Type of task: Supervised, single-label heartbeat classification. It is multiclass classification if the dataset has more than two heartbeat labels, or binary classification if it has exactly two. This follows the classification framework in Tan et al. (2005).

The objective is to develop a compact supervised machine-learning model that takes a normalized, fixed-length ECG heartbeat segments as input and predicts its heartbeat category, while maintaining acceptable classification performance after quantization and synthesis for deployment on an embedded device (Moody & Mark, 2005).

1.3 Project objectives

The technical objectives below overview the data-mining objective (heartbeat classification) into concrete, testable goals for a MAX78002 system (Analog Devices, n.d.). Every objective is stated in order for possible verification during evaluation.

Classify heartbeat types: Develop a model to distinguish the heartbeat categories agreed upon. Finalize category definitions and decide how unclassified beats will be handled before any model training begins (Silva et al. 2025).

Generalize to unseen recordings: To assess performance on ECG recordings not represented in training. This is to check whether the model can classify heartbeats from ECG recordings it never encountered during training.

Evaluate robustness to ECG noise: Measure performance on clean signals and on noisy versions of the test signals. This evaluation is to check if the model still classifies heartbeats correctly when the ECG signal is noisy (Alhalabi et al. 2025).

Develop a compact model: Choosing architecture that will keep model size and computational demands suitable for the target workflow and at the same time keep measure of classification performance. Model complexity and evaluation are relevant considerations in the classification framework described by Tan, Steinbach, and Kumar (2005).

Establish MAX78002 workflow: To train and evaluate the model, apply the MAX78002-compatible quantization and synthesis workflow, then generate the device-specific outputs. Quantization makes the numbers smaller and hence simpler for the chip to use. Synthesis organizes the layers into device-specific files (Analog Devices Inc, n.d.).

Assess quantization impact: To compare and document the model's classification metrics before and after quantization on the same data.

Report performance by class: To evaluate overall and class performance. This is to show how well the model recognizes each heartbeat type.

1.4 Success criterias

The thresholds below are merely proposed minimum targets and should be finalized after the dataset and class distribution are confirmed.

Table 1

Measurable success criterias

Criteria	***Measurable success criteria***
Predictive performance	The model performance meets the established acceptance target and isn't considered solely by accuracy. A study on ECG classification reported several measures of performance instead of relying on a single one (Issa et al. 2023).
Minority-class performance	The model's detection of each of the target minority classes at an adequate level is decided before final testing (Silva et al., Table 9., p 24, 2026).
Noise robustness	The model must be evaluated using controlled noise added to the test signals (the ECG). Degrading of performance under specified noise conditions remains with the established tolerance (Alhalabi et al. 2025).
Model complexity	For the model to stay compact and to fit within the resource budget for the intended deployment. The resource measures of model size and inference cost can be reported (Analog Devices, n.d.).
MAX78002 compatibility	The successful completion of the workflow, that the designed application fits the target resource budget, and synthesis doesn't cause an unsatisfactory change in prediction.

The project will be treated as successful when all five criteria pass and the dataset split, preprocessing, model configuration, quantization settings, and evaluation results are documented well enough to reproduce. The use of explicit evaluation measures and model-complexity checks is consistent with the classification and evaluation framework discussed by Tan et al. (2005).

1.5 Constraints

A constraint here is defined as a limitation we cannot remove, only design around. Each is expressed as (i) what is fixed, (ii) the measurable form which limits the project, and (iii) the forced design consequences.

Since the MAX78002 has memory limitations for neural-network weights, input data, and output buffers, large or highly connected models may fail during synthesis even if they achieve good classification accuracy in a desktop environment. Therefore the synthesis report will need to confirm the architecture fits within the finite on-chip resources (Analog Devices, n.d.).

If a dataset is chosen that has significantly more normal than abnormal beats, it is possible that a model could achieve strong overall accuracy while resulting in poor performance on

rare classes. Training and evaluation will need to account for the class distribution as well as class-specific metrics, and avoid relying on accuracy alone. For example, MIT-BIH heartbeat classes are reported as imbalanced, with the normal class substantially more frequent than several other classes. Tan et al. (2005) also discuss evaluation challenges associated with imbalanced classes, and model complexity and training size are relevant considerations in the data-mining framework described.

> *The training, tuning, and repeated experiments are in our minds limited by the host computer's available processing power, memory, and time. A compact architecture and reproducible experiment plan will help keep our development feasible. The MAX78002 itself is the inference target for us, not a substitute for the development computer used to train the model. Without a physical hardware kit, our project cannot verify actual board operation, timing, power consumption, or behavior with a physical ECG input. However, it will be able to demonstrate software-level quantization and synthesis. Checks should therefore be made by us, so that the generated model fits reported device resources, and evaluate the model using software, but these checks do not establish measured on-device performance.*

1.6 Risks and challenges

Patient-level data leakage could inflate results. Heartbeats from the same person can share individual signal characteristics. In the case that beats from one person appear in the test set and the training set, during test performance there may be an overstate in how well the model can handle unseen people. A mitigation could be to split data by patient or recording before segmenting beats and by that keep the test set untouched up until the evaluation. This follows the inter-patient evaluation approach discussed in ECG-classification research and supports the careful model evaluation emphasized by Tan et al. (2005).

Class imbalance might potentially hide poor detection of rarer heartbeat types. A classifier may score well overall by favoring the most common class while not picking up minority classes. This issue is well documented in ECG heartbeat datasets. A mitigation could be to measure class counts before training. If one considers class-weighted loss within the training data only and also if one evaluates the recall and precision of each class - this should keep a natural class distribution of the test set (Silva et al. 2025).

Since there is no physical MAX78002 kit that means hardware operation cannot be confirmed. Software synthesis and compilation has the ability to demonstrate workflow compatibility, but cannot verify actual, timing, power consumption, or ECG input integration. A mitigation could be to document successful quantization, synthesis, and code compilation. Furthermore, one should make no claim of physical-device validation.

The noise in actual ECG signals may differ from the one in training data. A model evaluated on relatively clean ECG may perform worse when electrode noise or inaccurate beat segmentation alter the waveform. A mitigation could be to apply ECG-only filtering and normalization. Because there is no physical kit, this will not validate robustness to every real-world acquisition condition (Lee et al., 2012).

Access to limited training and experimentation resources such as restricted CPU, GPU, RAM, or storage can reduce the number of experiments and make searches impractical. This could prevent the discovery of a better model or make results difficult to reproduce. A mitigation could be to use a reproducible experiment configuration, compact baseline models and controlled experimentation rather than unrestricted trial and error.

2. Data Understanding

2.1 Dataset Description

The dataset used in this project is the *MIT-BIH Arrhythmia Database*. It consists of 48 records from 47 subjects. Each record contains approximately 30 minutes of two-channel ambulatory ECG recordings. The signals were digitized at a sampling frequency of 360 samples per second per channel, with an 11-bit resolution over a 10 mV range. The recordings were collected by the BIH Arrhythmia Laboratory between 1975 and 1979. Twenty-three recordings were randomly selected from a collection of approximately 4´000 24-hour ambulatory ECG recordings. The remaining 25 recordings were selected specifically to include less common but clinically significant cardiac arrhythmias that would not be well represented in a random sample. Each record was independently annotated by two or more cardiologists. Disagreements were resolved to produce computer-readable reference annotations for each heartbeat. The database contains approximately 110´000 heartbeat annotations in total (Moody & Mark, 2005).

2.2 Target Classes

The heartbeats are to be classified according to the target classes. The target classes are derived from the original heartbeat annotations provided in the MIT-BIH Arrhythmia Database. These annotations were produced by two or more cardiologists. The original annotation types are mapped to the four target classes according to their corresponding cardiac rhythm category. Annotations that do not correspond to any of the four target classes are excluded from the classification entirely (Moody & Mark, 2005) (see Figure 1).

Table 2

Mapping of MIT-BIH annotations to the four target classes

Target Class	***Annotation (Symbol)***
Normal type	N
Supraventricular ectopic	A
Ventricular ectopic	V
Fusion	F

2.2.1 ECG Signal and Heartbeat Representation

An electrocardiogram (ECG) records the electrical activity of the heart over time. A typical ECG waveform consists of characteristic components, including the P wave, QRS complex, and T wave (see Figure 1). The P wave represents atrial depolarization, the QRS complex represents ventricular depolarization, and the T wave represents ventricular repolarization. The R wave is the prominent positive deflection within the QRS complex and reflects ventricular depolarization (Merck Manual Professional Edition, n.d.). In this study, the annotated R-peak is used as a temporal reference for identifying and extracting individual heartbeat segments from the continuous ECG recordings.

Figure 1
Typical ECG waveform from recording

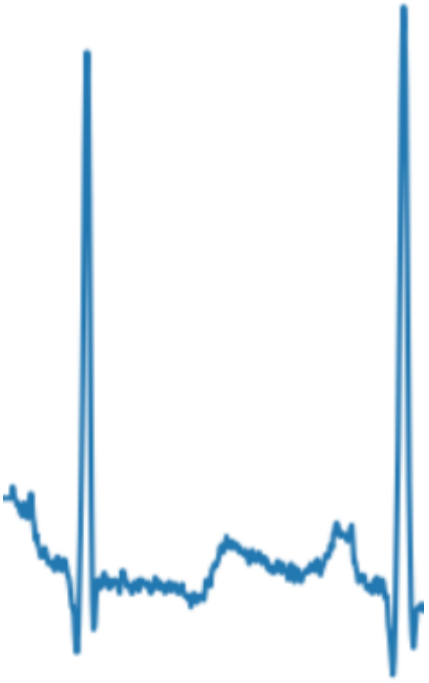
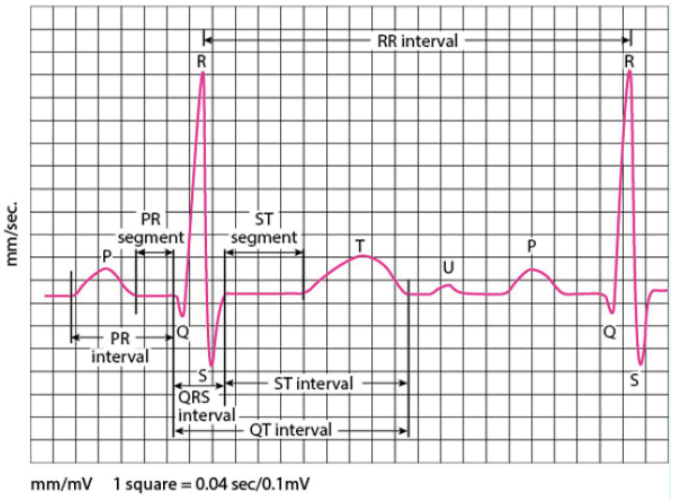


Figure 2
Heartbeat representation



2.3 Exploratory Data Analysis (EDA)

To understand the structure and characteristics of the ECG data before model development, several exploratory visualizations and data analyses were conducted. This section outlines the key visual and statistical insights extracted from the dataset.

2.3.1 Example ECG Signal

To examine the general morphology of the raw signal, a representative 5.5 second segment of record 100 (channel 1, MLII lead) was plotted (see Figure 3). To confirm that the provided beat annotations correctly align with the QRS complexes, the annotated R-peak were overlaid on the same segment of record 100 (see Figure 4).

Figure 3

Record 100 - first ~5.5 seconds (channel 1)

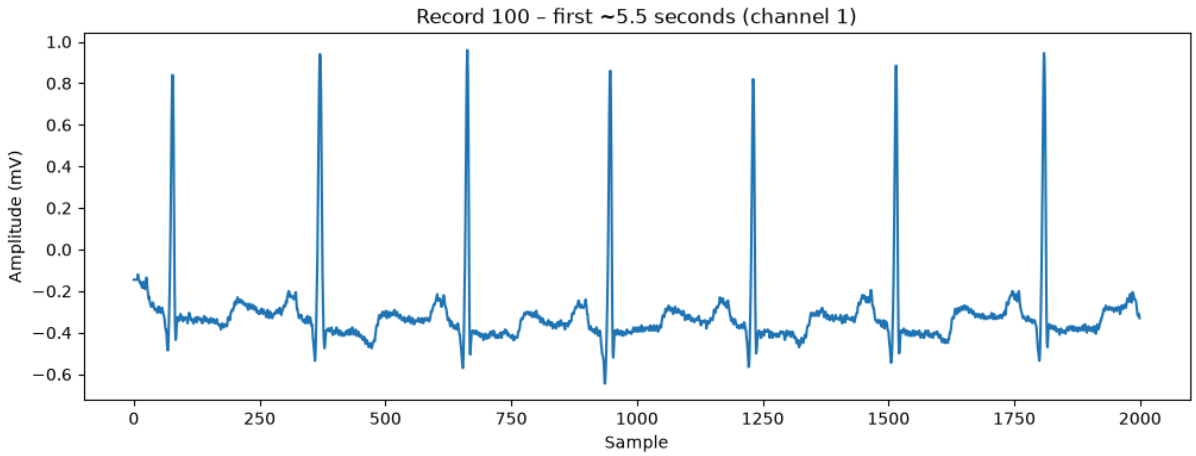
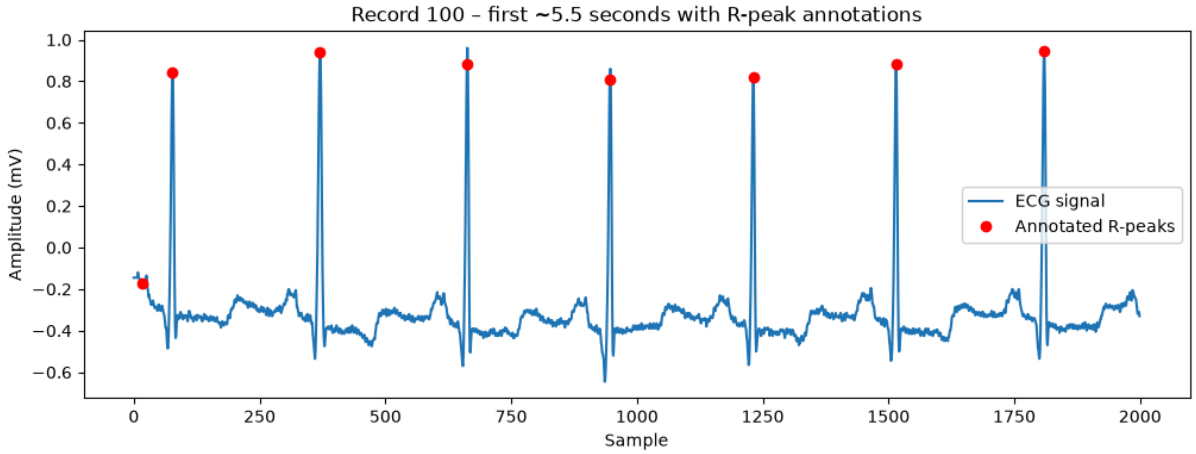


Figure 4

Record 100 - first ~5.5 seconds with R-peak annotations (channel 1)



Furthermore, to inspect the shape of individual heartbeats in isolation, a single beat was extracted from record 100 using a fixed window of 100 samples before and 200 samples after the annotated R-peak (see Figure 5). This represented a full P-QRS-T morphology from class N. To visually characterize each target class, representative beat examples were extracted using the same fixed window. The selection of records was based on official PhysioNet beat-type tables. Record 200 represents the V-class (826 ventricular beats), record 209 the S-class (383 atrial premature beats), and record 213 the F-class (362 fusion beats) (Moody & Mark, 2005). Figure 6 displays these representative isolated beats side by side, while Figure 7 illustrates longer multi-beat segments for each class to show how they appear within their surrounding rhythms.

Figure 5

Isolated single heartbeat (Record 100, class N)

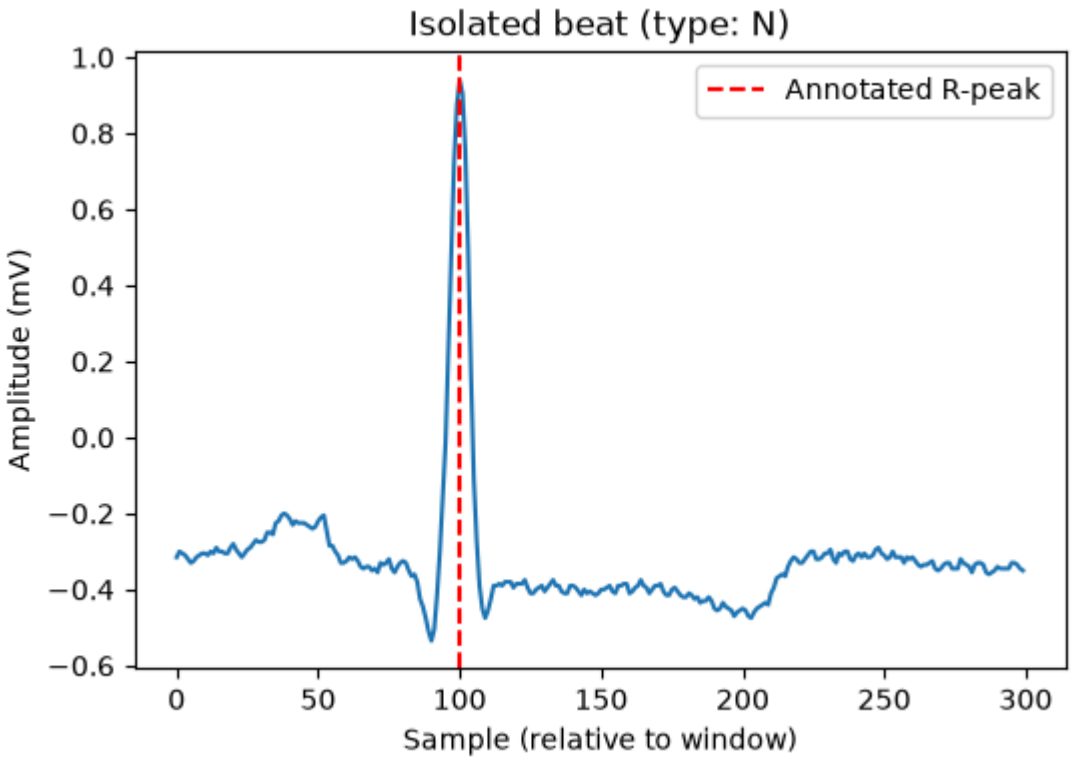


Figure 6

Isolated heartbeat examples from all four target classes

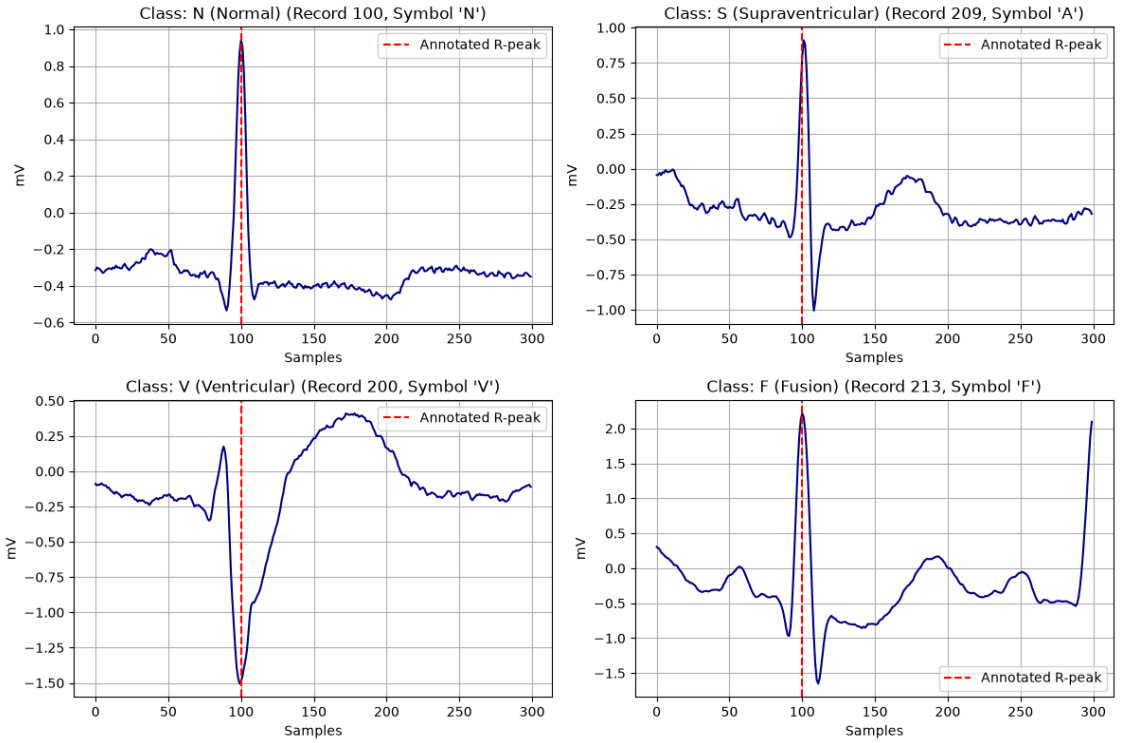
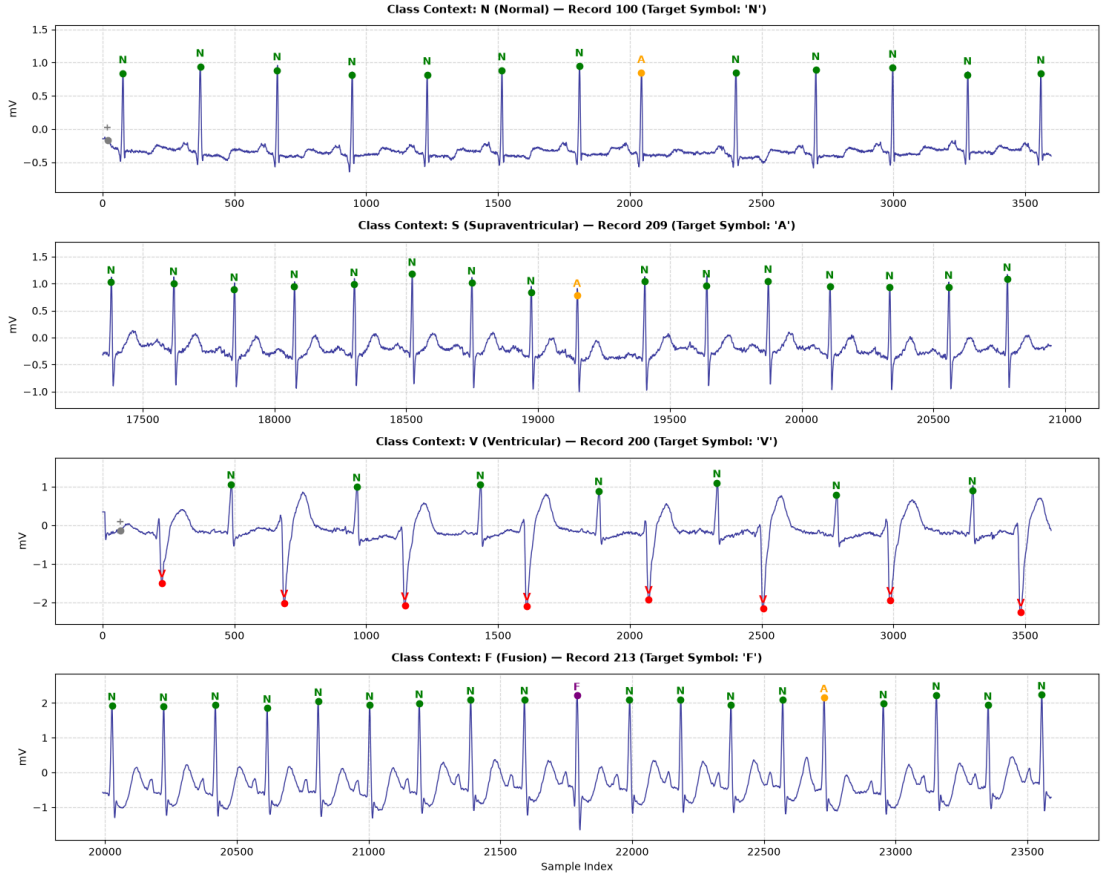


Figure 7

Example ECG-signal from four classes

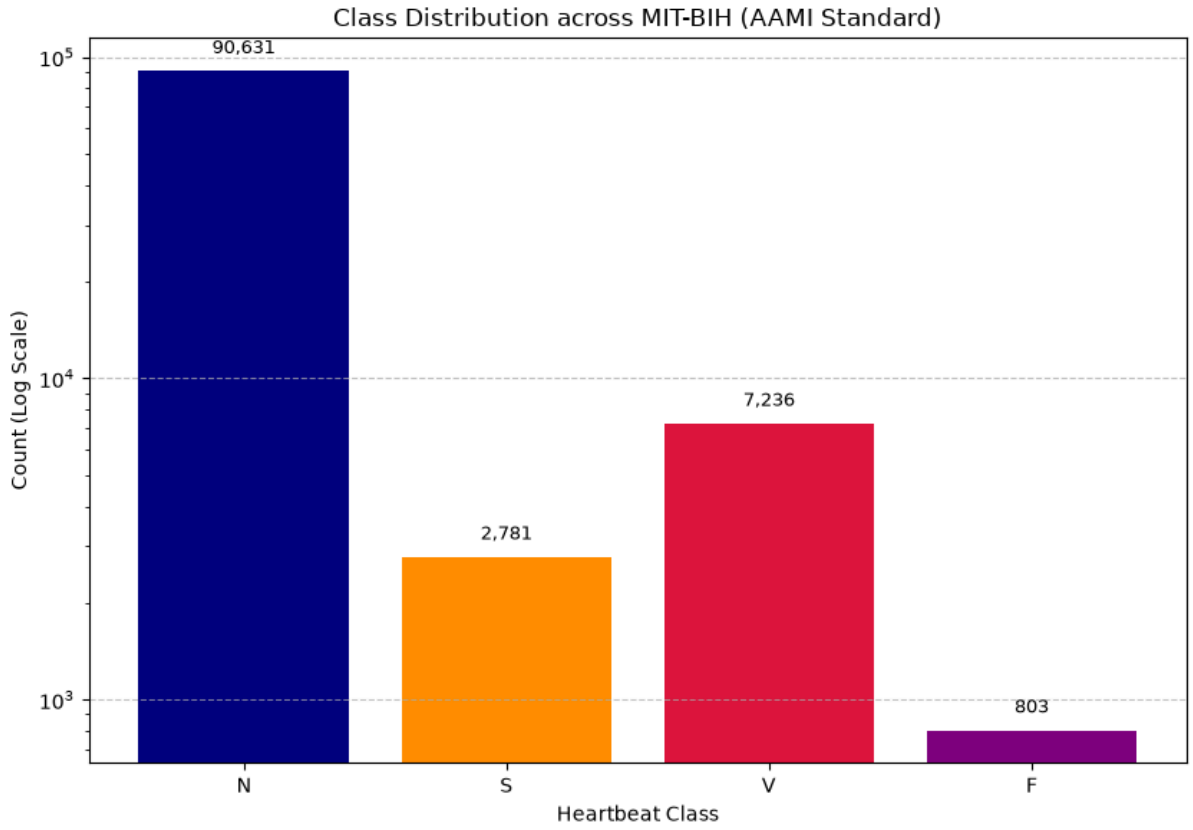


2.3.2 Class Distribution

The distribution of the four target classes across the entire dataset was computed by mapping the original MIT-BIH annotation symbols to the AAMI-standard classes (N, S, V, F) and counting occurrences across all 48 records (see Figure 8). The dataset is strongly imbalanced. Class N accounts for 89.33% of all beats (90,631 beats), while F represents only 0.79% (803 beats). In addition, classes S and V make up 2.74% (2,781 beats) and 7.13% (7,236 beats), respectively.

Figure 8

Class Distribution

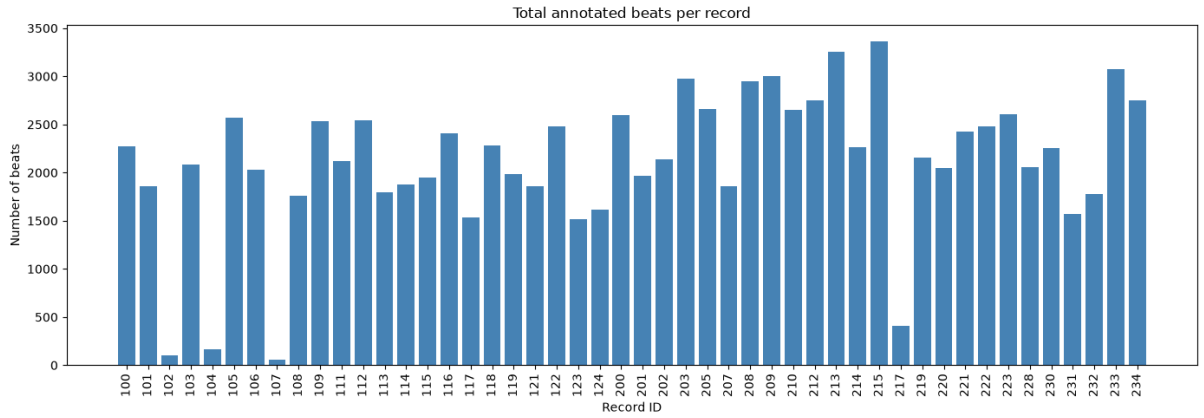


2.3.3 Record-Level and Subject Distribution Variability

The number of annotated beats also varies substantially across records, as illustrated in the bar chart in Figure 9. As can be seen in the dataset documentation, several records are heavily dominated by one particular beat type. For instance, record 232 is predominantly composed of S-class beats (1 382 out of 1780 total), in contrast to the dataset wide pattern where class N is dominant. Similarly, the F class is heavily concentrated in a small number of records, with record 208 alone accounting for 373 of the 803 total F-class beats (Moody & Mark, 2005).

Figure 9

Total annotated beats per record

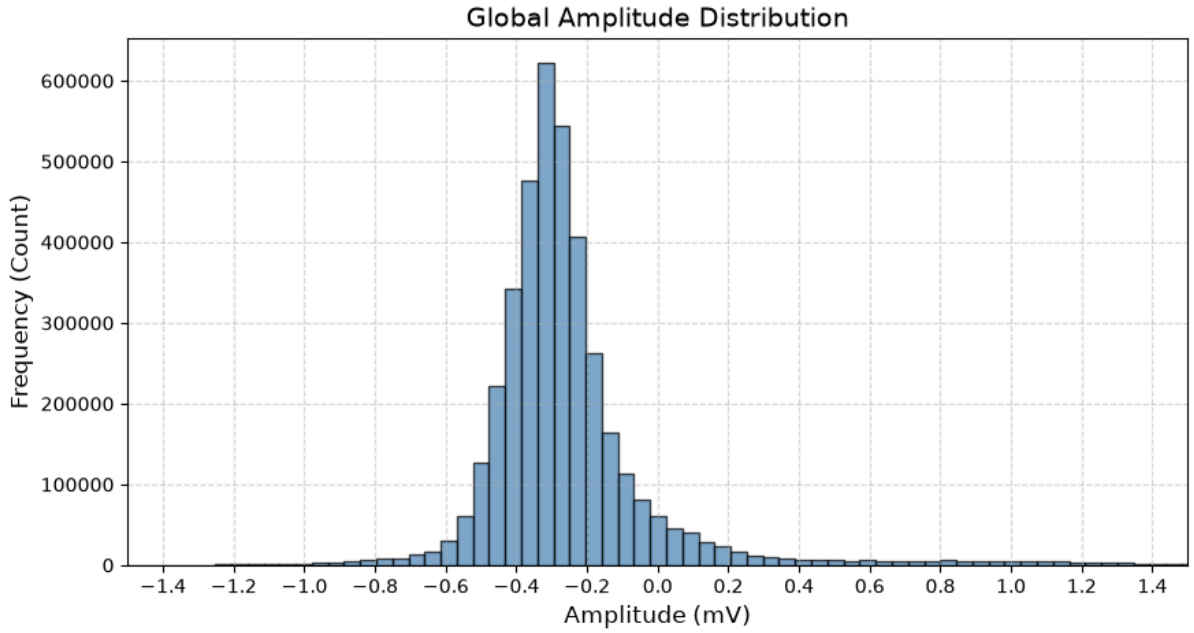


2.3.4 Signal Amplitude Distribution

To examine the overall range of signal values, a histogram of all amplitudes across the entire dataset from channel 1 was computed and is illustrated in Figure 10. The distribution peak around -0.3 mV reflects the baseline resting amplitude between heartbeats, combined by a long tail extending toward larger, less frequent amplitude values produced by the QRS complex. Most values fall within approximately ±1.0 mV, though some outliers extend further.

Figure 10

Amplitude distribution



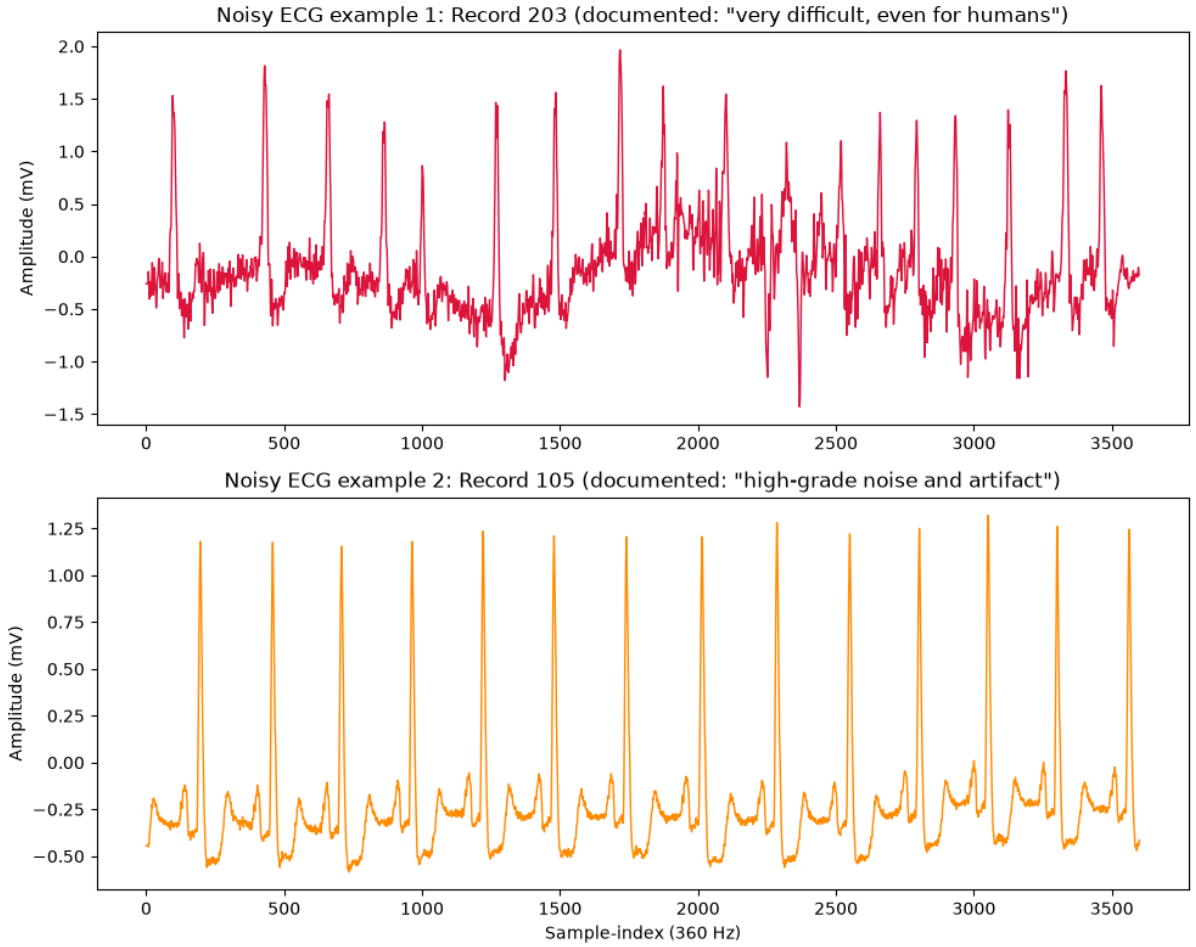
2.3.5 Examples of noisy ECG

Two records were selected based on the official PhysioNet documentation to illustrate noise present in the dataset. Record 203 was selected considering that the official documentation describes it as containing "considerable noise in both channels, including muscle artifact and

baseline shifts... a very difficult record, even for humans". To illustrate noisy examples further, record 105 was chosen based on the fact that the documentation pointed it out as "the predominant feature of this tape is high-grade noise and artifact" (Moody & Mark, 2005). Figure 11 shows a 10-second segment from each record, illustrating the irregular, high-frequency disturbances compared to the clean signal shown in Figure 3 (see 2.3.1 Example ECG Signal).

Figure 11

Two examples of noisy ECG signals



2.4 Data Quality

2.4.1 Missing Data

All records were checked for missing files, failed loading, missing signal values (NaN), invalid or empty annotation symbols, and gaps between consecutive annotated beats. The results demonstrate successful loading with all expected files (.dat, .hea, .atr) present for all 48 records. Furthermore, no NaN values were found in any signal, and no empty or invalid annotation symbols were present, and also no records contained zero annotations. The largest gap between consecutive annotated beats across the dataset was 5.87 seconds, found in record 232, which was noticeably larger than in the remaining records.

2.4.2 Extreme Amplitudes

For each record in channel 1, the minimum, maximum, and standard deviation of the signal amplitudes were determined. Maximum values ranged from 0.71 mV to 5.12 mV, and minimum values ranged from -5.12 mV to -1.12 mV. The standard deviation ranged from 0.165 mV to 0.858 mV. The most extreme values were found in record 116, with a maximum of 5.115 mV and a minimum of -5.120 mV.

2.4.3 Corrupted Segments and Class Imbalance

To find corrupted segments, a definition of what is meant by a corrupted segment was first established. A record was considered potentially corrupted if its number of samples differed from the other records, or if the data in either channel did not match the declared signal length. This definition addresses the overall structural integrity of each record, rather than local artifacts such as flat segments or electrode disconnections within an otherwise intact signal. All 48 records contained exactly 650,000 samples, and both channels were complete in every record. No record was flagged as corrupted under this definition.

The class distribution across all 48 records is highly imbalanced (Figure 6, Section 2.3.2). Of the 101,451 beats belonging to the four target classes, class N accounts for 89.33% (90,631 beats), followed by V with 7.13% (7,236), S with 2.74% (2,781), and F with 0.79% (803).

2.4.4 Differences Between Recordings and ECG Channels

The records were compared in terms of the total number of beats, the number of beats in each of the four classes, and signal amplitude. The number of beats per record varied and is visualized in Figure 7, with the lowest count in record 107 (59 beats) and the highest in record 215 (3 363 beats). The class composition also differed between records. For example, record 232 consisted mostly of class S beats (1,382 of 1,780, 78%), and record 107 contained only class V beats (59 of 59). Class F was concentrated in a few records, with record 208 contributing 373 of the 803 class F beats (46%).

As mentioned earlier (Section 2.1), the ECG recording of each record was done using two channels.To compare the two channels of each record, the channel names were extracted from the header file of each record, which showed six different channel combinations. The most common combination was (MLII, V1), found in 40 of the 48 records. The combinations (MLII, V5), (V5, V2) and (MLII, V2) each occurred in 2 records, and (V5, MLII) and (MLII, V4) in 1 record each. Lead MLII was present in 46 of the 48 records, V1 in 40, V5 in 5, V2 in 4, and V4 in 1. MLII was the first channel in 45 records and the second channel in 1 record (record 114). Two records, 102 and 104, did not contain MLII.

2.4.5 Possible Noisy Regions

To identify possible noisy regions, the absolute difference between consecutive samples was computed for each channel of each record. Samples where this difference exceeded the mean plus three standard deviations (computed per record and channel) were flagged. The number of flagged points per record and channel ranged from 3,707 (record 200, channel 2) to 29,253 (record 213, channel 1), and 1,669,564 points were flagged in total across all

records and channels. The threshold values ranged from 0.031 mV (record 101, channel 2) to 0.343 mV (record 116, channel 1).

2.5 Leakage Analysis

Data leakage constitutes a potential methodological concern in this study due to the structure of the MIT-BIH Arrhythmia Database where multiple heartbeat annotations originate from the same subject and recording. A random split at the individual heartbeat level would allow heartbeats from the same recording to occur in both the training and test sets. This may lead to overly optimistic performance estimates as the model could exploit recording-, and subject-specific characteristics rather than learning features that generalize across different subjects.

Such characteristics may include ECG morphology, baseline amplitude, heart rate, noise patterns, and other subject-specific signal properties.

Furthermore, adjacent heartbeats within the same recording are not statistically independent and may exhibit highly similar characteristics. A heartbeat-level split would therefore compromise the independence of the test set and potentially inflate the estimated generalization performance.

To mitigate this risk, all heartbeats originating from the same subject are assigned exclusively to either the training, validation, or test set. This ensures that no subject-specific information is shared across partitions and that the test set represents previously unseen subjects. The resulting evaluation therefore provides a more reliable estimate of the model's ability to generalize to unseen patients.

3. Data Preparation

3.1 Record Selection

Realistically the only thing that matters is that we do not use the same patient recordings for both training and testing. If we were to divide the data like most data where we would all the samples shuffle them and randomly distribute them over training and testing in a 80/20 split, the model would have heart beats from the same patient in both training and testing pile leading to a network that instead of seeing pattern in the heartbeats instead learns specific patients. This would then lead to a really good accuracy but would be useless for new cases. So instead we will follow the recommendations from the task description where we divide into the known to be working DS1 and DS2 split. Where DS1 is for training and validating while DS2 is for testing. (Silva et al., 2025)

So, for DS1 (training and validation):
101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, 230

and for DS2 (testing):

100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234

The excluded recordings are due to a very low beat count however we could actually still use them if we wanted since we are not looking at the time between beats and thus extract the beats as separate windows of a fixed length.

3.2 Channel Selection

Channel combinations across all records:

('MLII', 'V1'): 40 records
('MLII', 'V5'): 2 records
('V5', 'V2'): 2 records
('MLII', 'V2'): 2 records
('V5', 'MLII'): 1 records
('MLII', 'V4'): 1 records

Looking at the different combinations we can see that all but 3 have ”MLII” as channel 1. After closer inspection we can see that:

102 ('V5', 'V2')
104 ('V5', 'V2')
114 ('V5', 'MLII')

102 and 104 are the ones that have missing MLII channels and are already excluded from the training and testing set. 114 on the other hand is being used and has MLII as its second channel (index 1) meaning we just need to avoid hardcoding MLII as index 0 and instead look for the name.

We will be using MLII mainly because it is the most consistent. As we saw it is only missing from two of the patients. As for how to handle missing data, in this case we can simply just not use them since it is very few (2 out of 48) and they are already excluded from the DS1 and DS2 data split and mixing different viewpoints would only make the model worse. However, if it would be the case where we had multiple we would have to compromise by either not using a lot of the data, some sort of normalizing or view point adapting or training two separate models for each viewpoint.

3.3 Beat Extraction

As for the beat extraction the main problem is that we want a set window length that captures enough features so we can make a prediction from it. The paper (in foot note) reduced the beats into 4 features that can be used. They reduced it from 31 features by basically doing a trial and error where they started off with one feature and tested the result then added and another and compared, they did this until they had the most optimal combination of features. These 4 features are R-peak amplitude, QRS fourier coefficient, T-wave duration and pre-RR interval. The R-peak is how tall the main spike of the heart beat is. The QRS fourier coefficient which basically is when we break down a single signal into sin curves and the maximum is the one that has the highest frequency (simply put it is a single value describing the shape of the peak). The T-wave describes how long the recovery is after the peak. And finally the pre-RR interval which is how fast the beat was compared to

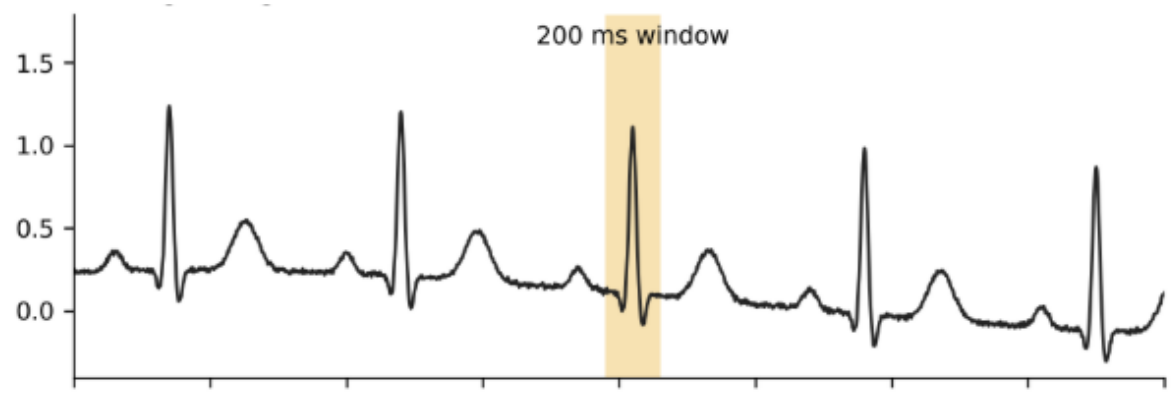
the other normal beats before. It is worth noting that this paper did not implement a 1D CNN like we are going to do. The features will however stay the same but since we are working with single beats with a fixed window length we can't actually work with the pre-RR interval since we will have every beat separated. (in the paper they used it because they used an ANN). Although we might implement something like an ANN or SNN after the recommended CNN so features might change a bit. (Lakshminarayan & Basil, 2016)

With this said, the R notations where the peaks are we get from the dataset itself in annotations. And since there are no exact markings for where beats start and end in the dataset we will have to form them around the R-peak. Meaning we will get “peak - before” and “peak + after” and together this will create the full window length. The exact window size is 90 samples before and 144 samples after so we get the 3 features we need to classify each beat.

3.4 Signal Preprocessing

Since everyone has different hearts their “normal” heart beats will look different, this would not work for a model where we separate every beat. Meaning first we need to make sure that every beat looks somewhat the same (has the same baseline). This will also help with a slow drift in the heart beat, which is something we see in the paper and is rather hard to see in our visualized samples but will be easier to see with longer samples from the recording. So we take 200ms centered around the R peak and take the median giving:

Figure 12
R peak 200ms centered window



After flattening the peaks we expand the filter to 600ms to also capture the T-wave and after taking the median of this 600ms centered around the R-peak we get:

Figure 13

Flattened R peak & filter expansion

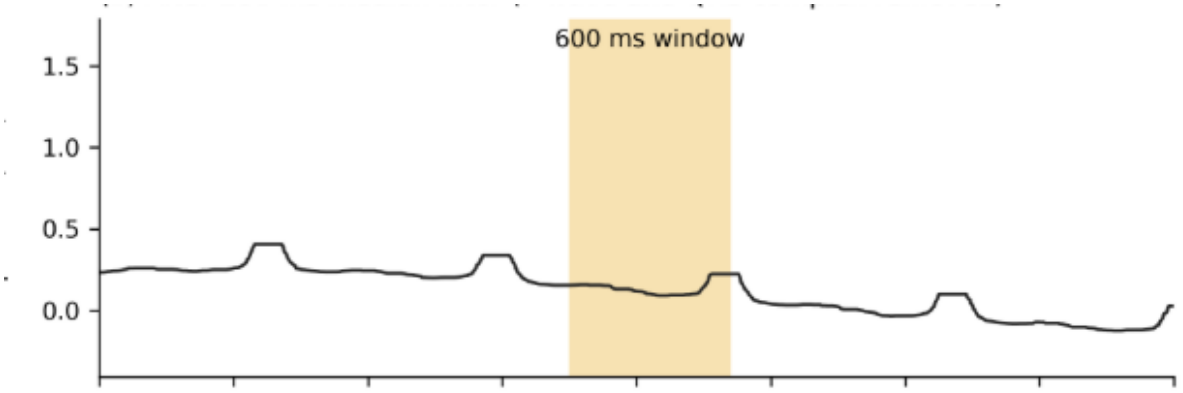
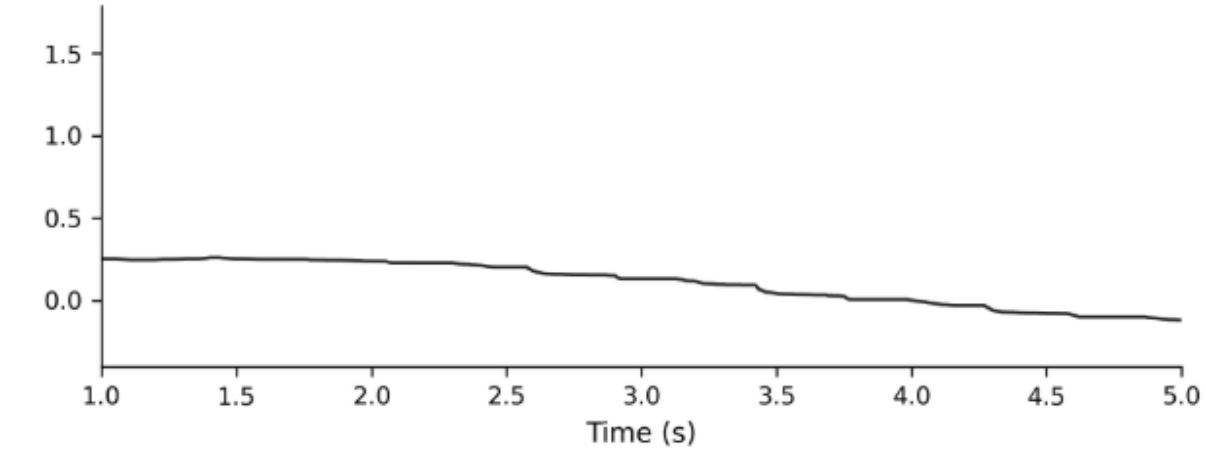


Figure 14

Totally leveled signal



The median we are left with is the drift that we then subtract from the original leaving us with a totally leveled signal that has a united base. So if we do this for each beat all beats are on the same baseline and don't have any drift. Note heart beats can still be different heights they are just on the same baseline. The paper then downsampled for its ANN approach so it would be less demanding on the hardwear, this should not be needed for our CNN approach but is worth noting in case we change model type later or test other models. Furthermore they bring up the need for a notch filter and use the size of 2Hz. This is done since there are "powerline" nice at 60Hz, hard to tell if this is needed for us but there are very little actual beat data at 60Hz so it will not break anything if we use it. (Lakshminarayan & Basil, 2016)

We now extract each individual beat where we use our fixed window centers at each R-peak. Also where we remove the samples where we are missing labels. And in some cases we will have to remove the first and last beat as well since they might not have the expected length for our set window. And finally when we have our individual beats we just want to normalize them so that we don't get some beats with much greater peaks leading to heavier weight change in training. This we can do in some ways but it seems getting the z-score is the most common and is where we basically make the beat have mean 0 and a standard deviation 1. We take each sample in the beat and do (X_i - mean) / standard deviation. (Benmessaoud et al., 2024)

(Patro & Sahu, 2015)

3.5 Label preparation

Class N: N, L, R, e, J
Class S: A, a, j, S
Class V: V, E
Class F: F

3.6 Class Imbalance

Since we have a very large difference between the number of beats within the classes we need to make sure we avoid a model that always just predicts N and gets a 90% accuracy. We will be doing this with class weights and balanced sampling. This means we are going to make sure that model gets a larger punishment for having the rarer classes predicted wrong and we are also going to make sure that we have V and F labels in both training and validation sets. We do not get a validation set with 0 of those classes. If we did not do this and had 0 F or V labels in validation the model would never get punished for not knowing them and if we do not have the weights it will get stuck at predicting N all the time. We are taking the weight and rescaling them with (total samples / number of classes * counts of this class) and for the balanced sampling (1 / counts of this class)

3.7 Augmentation

As we are normalizing the amplitudes and the baseline wander there is no real point in adding them as augmentations since that would just mean we did not need to normalize them in the first place. However, adding some temporal shifting where we shift the window a bit and pad the edges, as well as adding some noise could prove useful for our model. Note that after we have added noise we will have to renormalize the signal so that we keep the mean 0 and std 1 format. We are now not renormalizing after since we are normalizing over entire recordings the first time meaning if we do it again we get one normalization on record level and one on window level.

3.8 Final Dataset

We do DS1 and DS2 split and within the DS1 split we make sure that the 2 records where we have F and V get some samples in both validation and training so we can actually validate for those classes as well as train for them. We can make sure we get 108,114,201,207,223 in validation to make sure we can actually test for V, F while training.

3.9 Reproducibility

Since we are working with a fixed split into train, validation and test set we will always get the exact same split when we run the script file. However the things that are random are the augmentation we apply to the test data. As for our software versions we use python 3.12 and the latest versions on pytorch and numpy. Some fixed parameters from our config file include:

200ms and 600ms filter median normalization. (0.2 * 360 and 0.6 * 360 = 72 and 216 samples)
Windows size (input) 90 samples before and 144 after to catch our 3 features.
Augmentation: Shift +-5 and epi 0.1. Random Seed: 0.

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